

Adaptive Learning Systems for Mathematics Education

Abstract

This paper reports on the deployment of an adaptive learning system in high-school mathematics classrooms. The system p

Methods

We deployed the system in 8 classrooms with 240 students over one academic year. Pre- and post-tests measured mathem

Results

Students using the adaptive system showed a mean gain of 18.5 points ($SD=4.2$) on the standardized post-test, compared t

Discussion

Adaptive learning systems can produce measurable gains in mathematics achievement. The pedagogical mechanism appe

Limitations

The study used a quasi-experimental design without random assignment. Selection effects cannot be fully ruled out.